

pkshatech / **simcse-ja-bert-base-clcmlp**

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Sentence Similarity



sentence-transformers



PyTorch



Transformers



Japanese

bert

feature-extraction

arxiv:2104.08821



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Train

Use this model



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🌐 Sentence Similarity

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🔗 Japanese SimCSE (BERT-base)

[日本語のREADME/Japanese README](#)

🔗 summary

model name: pkshatech/simcse-ja-bert-base-clcmlp

This is a Japanese SimCSE model. You can easily extract sentence embedding representations from Japanese sentences. This model is based on c1-tohoku/bert-base-japanese-v2 and trained on JSNLI dataset, which is a Japanese natural language inference dataset.

🔗 Usage (Sentence-Transformers)

You can use this model easily with sentence-transformers.

You need fugashi and unidic-lite for tokenization.

Please install sentence-transformers, fugashi, and unidic-lite with pip as follows:

```
pip install -U fugashi[unidic-lite] sentence-transformers
```

You can load the model and convert sentences to dense vectors as follows:

```
from sentence_transformers import SentenceTransformer
sentences = [
    "PKSHA Technologyは機械学習/深層学習技術に関わるアルゴリズムソリューションを展開している。",
    "この深層学習モデルはPKSHA Technologyによって学習され、公開された。",
    "広目天は、仏教における四天王の一尊であり、サンスクリット語の「種々の眼をした者」を名前の由来
]

model = SentenceTransformer('pkshatech/simcse-ja-bert-base-clcm1p')
embeddings = model.encode(sentences)
print(embeddings)
```

Since the loss function used during training is cosine similarity, we recommend using cosine similarity for downstream tasks.

[🔗 Model Detail](#)

[🔗 Tokenization](#)

We use the same tokenizer as `tohoku/bert-base-japanese-v2`. Please see the [README of tohoku/bert-base-japanese-v2](#) for details.

[🔗 Training](#)

We set `tohoku/bert-base-japanese-v2` as the initial value and trained it on the train set of [JSNLI](#). We trained 20 epochs and published the checkpoint of the model with the highest Spearman's correlation coefficient on the validation set [^1] of the train set of [JSTS](#)

[🔗 Training Parameters](#)

Parameter	Value
pooling_strategy	[CLS] -> single fully-connected layer
max_seq_length	128
with hard negative	true
temperature of contrastive loss	0.05
Batch size	200
Learning rate	1e-5
Weight decay	0.01
Max gradient norm	1.0
Warmup steps	2012
Scheduler	WarmupLinear
Epochs	20
Evaluation steps	250

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[^1]: When we trained this model, the test data of JGLUE was not released, so we used the dev set of JGLUE as a private evaluation data. Therefore, we selected the checkpoint on the train set of JGLUE insted of its dev set.